

HARIRAMAKRRISHNAN RAMACHANDRAN

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PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Designed, developed, and iterated on **prompt templates** for Large Language Models to achieve specific goals, such as simulating prompt injection and data elicitation for security testing.
- Experimented with and implemented advanced prompting methodologies including **few-shot prompting** and **chain-of-thought prompting** to maximize AI interaction efficiency and effectiveness.
- Identified failure modes and unexpected outputs in LLM behavior, addressing them through systematic **prompt refinement** and optimization, contributing to system safety.
- Collaborated with AI Researchers and Engineering teams to integrate effective prompts into AI security platforms and other advanced ML-based systems.
- Built and maintained an internal library of effective prompting strategies, techniques, and best practices, fostering knowledge sharing across engineering teams.
- Researched and applied **cutting-edge developments in LLMs** and AI security advancements to improve system performance and identify new adversarial challenges.
- Containerized prompt engineering tools and services using **Docker** and deployed them to an **AWS** environment, ensuring reliable execution and scaling.

SKILLS

AI/ML – Large Language Models (LLMs), Prompt Engineering, NLP, Generative AI, AI Security, Adversarial AI, Red-Teaming, Agentic AI Systems, LangChain, RAG, Vector Databases, Fine-tuning, Model Evaluation

Programming – Python, Java, SQL, Bash Scripting

Cloud & DevOps – AWS (EC2, S3, Lambda, CloudWatch), Docker, Kubernetes, CI/CD, Jenkins, Git

Data & Analytics – Data Analysis, Data Modelling, Data Warehousing, Power BI, Tableau, R, pandas, NumPy

Backend & APIs – Spring Boot, REST APIs, FastAPI, Microservices

Databases – PostgreSQL, MySQL, Oracle

PROJECTS

LLM Security Testing Framework

- Developed a Python-based framework to automate the generation and execution of adversarial prompts against various LLM APIs, simulating **prompt injection** and **jailbreaking** attempts.
- Integrated **LangChain** to manage prompt chains and context, enabling complex interaction scenarios for security vulnerability assessment.
- Implemented structured logging and analysis of LLM responses to identify patterns of failure and classify potential security risks, processing approximately 500 unique prompt variations per test cycle.

Customer-Facing AI Assistant Integration

- Designed and implemented a prompt engineering workflow for a customer-facing AI assistant, focusing on clear, concise, and secure interactions.
- Collaborated with a product team to gather requirements for AI assistant capabilities, translating business needs into effective **prompt designs** for core functionalities.
- Built a Python module to manage prompt variations and user context, ensuring consistent and reliable responses for an average of 150 daily user interactions.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)